

CO 2 - ASSESSMENT - 3

M. Phanish Chander
192425238
3TA0514 - Computer
Vision

(1) An image appears too bright. Explain and justify the most suitable gray level transformation to improve visibility, clearly describing the concept, application process and expected outcome. ①

(Ans) The most suitable grey level transformation for an overly bright image is negative Transformation or Intensity Reduction using Gamma Correction.

$$s = c \times r^{\gamma}$$

where,

r = input pixel intensity

s = output intensity

γ = correction value.

Application Process:

- i Read the image.
- ii Normalise pixel values b/w 0 to 1
- iii Apply gamma correction with $\gamma > 1$.
- iv Convert the image back to its original intensity range.

Expected outcome:

- i Excessive brightness is reduced.
- ii Hidden details become visible.
- iii Contrast improves.
- iv The image appears more natural and easier to interpret.

91
100

(2) A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

(Ans) A median filter is a non-linear filter that removes noise by replacing each pixel with the median value of its neighboring pixels.

Working Mechanism

(1) Select a window.

(2) Arrange all neighboring pixel values in ascending order.

(3) Replace the center pixel with the median value.

(4) Repeat for each pixel.

Impact on Noise Reduction

(1) Removes salt-and-pepper noise effectively.

(2) Preserves edges better than averaging filters.

(3) Produces a smoother image while maintaining important details.

(3) An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

(Ans) The most suitable sharpening method is Unsharp Masking. Unsharp masking sharpens an image by enhancing its high-frequency components (edges).

Working Principle :-

(1) Blur the original image.

(2) Subtract the blurred image from the original to obtain edge details.

(3) Add the edge information back to original image.

$$\text{Sharpened} = \text{Original} + k(\text{Original} - \text{Blurred}).$$

Expected Results:

- i Restore sharp edges.
- ii Enhance fine details.
- iii Improves image clarity.
- iv Makes object appear more distinct.

(2)

(4) Fine details in an image are not clearly visible. Describe and justify the filtered technique used to enhance details, including its effect on high-frequency components.

(Ans) A high-pass filter allows high-frequency components to pass while suppressing low-frequency components.

Working Principle:

- (1) Apply a high-pass filter mask.
- (2) Detect rapid intensity changes.
- (3) Enhance edges by emphasizing high-frequency information.

Effects on high frequency components:

- * Strengthens edges.
- * Enhances textures.
- * Makes fine details more visible.

Expected Results:

- * Better edge definition
- * Improved visibility of small structures
- * Increased image clarity

(5) A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

(Ans) The median filter is a non-linear spatial filter that removes impulse noise while preserving image edges.

Application Process :-

- (1) Choose a filter window.
- (2) Collect all neighboring pixels.
- (3) Sort the values in ascending order.
- (4) Replace the center pixel with median values.
- (5) Repeat the process for each pixel.

Expected Outcome :-

- i) Removes random noise.
- ii) Preserves edges.
- iii) Produces a smoother image.
- iv) Improves image quality (segmentation or feature extraction)

(6) An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

(801) The most suitable segmentation technique is Thresholding. Thresholding separates the object from the background using a threshold intensity value.

Working Principle :-

- (1) Select a threshold value T .
- (2) Compare every pixel intensity with T .
- (3) Pixels greater than T are classified as the object.
- (4) Pixels less than T are classified as the background.

$$g(x, y) = \begin{cases} 1, & f(x, y) \geq T \\ 0, & f(x, y) < T \end{cases}$$

Effectiveness:

- * Simple and fast
- * Clearly separates bright objects from dark backgrounds
- * Produces a binary image suitable for object detection and measurement.

- (9) You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.
- (10) Thresholding separates an image into foreground and background using a threshold value (T). Pixels brighter than the threshold are classified as the foreground, while the remaining pixels become the background.

Process:

- (1) Read the grayscale image.
- (2) Choose a threshold value (T).
- (3) Compare each pixel with T .
- (4) Pixels with intensity $\geq T$ become foreground (1).
- (5) Pixels with intensity $< T$ become background (0).

$$g(x, y) = \begin{cases} 1, & f(x, y) \geq T \\ 0, & f(x, y) < T \end{cases}$$

Expected Outcome:

- * Produces a binary image
- * Clearly separates the object from background
- * Suitable for image analysis and object detection.

(8) An image requires identification of object detection edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.

(9) The most suitable edge detection method is the Canny Edge detector. It identifies object boundaries while minimizing noise.

Working Process:

- (1) Smooth the image using a Gaussian filter.
- (2) Calculate intensity gradients.
- (3) Apply non-maximum suppression to thin the edges.
- (4) Use double thresholding to classify strong and weak edges.
- (5) Perform edge tracking by hysteresis to retain only meaningful edges.

Expected Outcome:

- * Thin and continuous edges.
- * Accurate boundary detection.
- * Reduced noise and false edge detection.

(9) Detected edges are noisy and fragmented. Describe and justify a method to refine edges and improve boundary detection, explaining its role in segmentation.

(10) The most suitable method is Morphological Operations, particularly Closing. Morphological operations process binary images using a structuring element to improve object boundaries.

Working Process :-

- (1) Apply Dilation to connect broken edges.
- (2) Apply Erosion to remove extra pixels while preserving connected boundaries.
- (3) This combination is called Closing.

Role in Segmentation :-

- * Connects fragmented edges
- * Remove small gaps.
- * Produces smooth and continuous object boundaries.
- * Improves segmentation accuracy.

Expected Outcome :-

- Continuous edges.
- Better object boundary detection.
- Reduced noise.

- (10) An application requires precise object boundary extraction for shape analysis. Explain and justify a suitable segmentation and boundary approach, describing its working and effectiveness.
- (20) The most suitable approach is Thresholding followed by Canny Edge Detection.

First, threshold separates the object from the background. Then, Canny edge detection accurately extracts the object's boundary.

Working Process :

- (1) Convert the image to grayscale.
- (2) Apply thresholding to segment the object.
- (3) Apply the Canny Edge Detector to identify object boundaries.
- (4) Refine boundaries using morphological operations if necessary.

Expected Outcome :

- * Accurate object segmentation.
- * Precise and continuous boundary extraction.
- * Suitable for shape measurement, recognition, and analysis.

Justification :

Combining thresholding with Canny edge detection provides reliable segmentation and highly accurate boundary extraction, making it ideal for shape analysis.